

Intelligent System for Plant Disease Detection and Remedy Recommendation

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Abstract

Plant diseases are a major problem in agriculture and they affect crop production every year. Many farmers cannot identify plant diseases at an early stage. Most of the disease detection process is still done manually by observing plant leaves, which is not always correct and also takes a lot of time. In many rural areas, expert help is not easily available, because of which diseases are detected late and crops get damaged. In this project, we have developed a web-based system for plant disease detection using deep learning models. In this system, the user first uploads an image of a plant leaf. The uploaded image is then checked by the system to find out whether the plant is healthy or affected by a disease. During the project work, several deep learning models were tried, including U-Net, ResNet50, Vision Transformer, VGG16, VGG19, AlexNet and DenseNet. For training and testing the models, the PlantVillage dataset was used, which includes images of both healthy leaves and leaves with different diseases. Some image preprocessing and data augmentation techniques were applied to improve the results. The results obtained during testing were satisfactory and showed that deep learning can be used for plant disease detection. The system also provides basic remedy suggestions after disease prediction. This project mainly aims to help farmers in early disease identification and reduce crop loss..

1 Introduction

Plant diseases represent a significant threat to agricultural output and food security in the world. The failure to identify the disease in good time, or in the right manner, has resulted in a very big percentage of wasted crop yield seeing as diseases are diagnosed at a later time, which is usually compounded by lack of information on the right treatment method. Most agricultural processes still involve manual observation in the detection of diseases, which is inefficient, subjective, and highly dependent on the expert knowledge of the individual. Another factor that prolongs diagnosis is the inability to access agricultural experts in time in rural areas where farmers do not have a solution that they can implement until crops are too damaged [1].

As machine learning methods continue to develop at a high rate, the automated detection of plant diseases has become a matter of growing interest. Initial work done by conventional machine learning techniques was heavily dependent on hand crafted features and was limited to generalization outside of the environmental settings of the original data [1,6]. Recent developments in deep learning, especially convolutional neural networks (CNNs) have overcome such constraints by learning individually discriminative spatial and texture-based features on raw images, and this leads to better classification [2].

A few CNN-based frameworks, including Slender-CNN have demonstrated good results on crops like corn and rice [5]. However, the model usually suffers performance loss when used in the real world due to factors such as background clutters, changes in illumination, and similarity between classes of a disease [3]. Image segmentation has become a vital preprocessing factor to curb such problems. Segmentation allows classification models to pay attention to disease-infested regions of the leaf when the leaf is in an intricate background, thus enhancing robustness and reliability of the classification models in real-life farm settings [6].

More recently, it has been proposed to use Vision Transformer (ViT) models that can also encode global contextual features using attention and supplement localized feature discovery of CNNs. New models that use convolutional and transformer based architecture as hybrids can be a key step towards precision agricultural application such as plant disease detection [4].

Irrespective of these technical developments, there is still a disparity in the process of translating high-performance detection models into accessible and practical tools

that end-users use. We suggest a smart web-based system based on automatic image preprocessing, leaf segmentation, and disease classification in the proposed work as a single pipeline. The proposed system maps the predicted disease labels to a structured remedy database in contrast with the previous studies which mostly concentrate on diagnosing the disease and thus farmers are left with quick treatment and prophylaxis suggestions. The system is trained using the PlantVillage dataset so that it can perform well in various crop species. The main aim of this paper is to fill the knowledge gap between the state of the art deep learning and real world applications in agriculture to facilitate the early detection of diseases and their prompt intervention.

1.1 Comparative Literature Review

S. No	Author / Title / Journal	Tools / Techniques / Dataset	Key Findings / Results	Limitations / Gaps Identified
1	Sunita Rani et al. – Comparative Analysis of Machine Learning Algorithms for Plant Leaf Disease Detection (2023)	Python (Scikit-learn), Random Forest, Naive Bayes, XGBoost, PlantVillage	ABOU achieved 94.1% accuracy; Naive Bayes performed weakly; preprocessing and augmentation improved results	Limited dataset diversity; not tested on real-field conditions
2	Shivani Singh et al. – Plant Leaf Disease Detection Using CNN Models: A Systematic Review (2022)	CNN Models (VGG, ResNet, EfficientNet, AlexNet), Segmentation (UNet, Mask R-CNN), Detection (YOLO, Faster R-CNN), TensorFlow, PyTorch, PlantVillage	High accuracy (98–99%); suggested lightweight CNNs for IoT/Drones	High computational requirements; dataset limitations
3	Neha Sharma et al. – Plant Leaf Disease Detection using ML, DL and Few-Shot Learning (2025)	Python, TensorFlow, PyTorch, Hyperspectral/Thermal Cameras, CNNs, Vision Transformers, FSL	Effective for small datasets; scalable AI; molecular diagnostics integrated	Limited crop diversity; requires specialized imaging tools

4	Murugavalli & Yopi – PLA-ViT: Vision Transformers for Precision Agriculture (2025)	Vision Transformers (ViT), Transfer Learning, IoT Sensors	Captures global features via self-attention; high accuracy; real-time monitoring	Heavy model size; may require high-end hardware
5	Baiju et al. – Slender-CNN for Corn, Rice & Wheat Crops (2025)	Slender-CNN, Multi-scale convolution, Image Processing Libraries	Lightweight model; suitable for farm-level deployment; multi-scale detection	Focused on three crops only; generalization unclear
6	Wubetu Barud Demilie – Plant Disease Detection and Classification Techniques: A Comparative Study (2024)	SVM, KNN, Random Forest, Naive Bayes, Decision Trees, CNN, ANN	Comparative study of ML and DL; highlighted real-time deployment challenges	Lack of experimental validation; mostly survey-based
7	Anh H. Vo et al. – Intelligent Recommendation System for Plant Disease Treatment (2023)	EfficientNet-B2, ResNet50, MobileNetV2, CNN, Transfer Learning, Cosine Similarity	Two-module system (detection + recommendation); accurate diagnosis	Needs real-field validation; remedy suggestions limited to dataset mapping
8	Emmanuel D. Escute & F. O. Irinkaye – Smartphone-Based Plant Disease Detection and Recommendation (2023)	CNN + ANN + KNN, Flutter/Android Studio, Firebase	Mobile-friendly system; real-time diagnosis and recommendation	Limited crops tested; dependent on smartphone camera quality

9	Anita Rani Mehta et al. – AI-Powered Innovations in Agriculture: A Systematic Review (2023)	Review of ML/DL: SVM, Random Forest, Decision Trees, CNN, ANN	Comprehensive analysis; identified research trends and future directions	No experimental validation; survey-based
10	Nishant Shelar et al. – Plant Disease Detection Using CNN (2023)	VGG-19 CNN, Keras, TensorFlow Lite	Achieved 95.6% accuracy; deployable on Android devices	Limited dataset size; may underperform in real-field conditions
11	Plant Leaf Disease Detection, Classification and Diagnosis: Computer Vision & AI Survey (2023)	CNN, SVM, KNN, Random Forest, ANN, MATLAB, Python, OpenCV	Hybrid ML/DL improves performance; transfer learning applied	Mostly survey-based; lacks real-time deployment analysis
12	Sammy Militante et al. – Plant Leaf Detection and Disease Recognition Using Deep Learning (2023)	CNN, ReLU, Pooling, Python Neural Network APIs, Data Augmentation, PlantVillage (35,000 images)	Achieved 96.5% accuracy across multiple crops; robust model	Dataset limited to PlantVillage; field performance not evaluated

2 Methods

2.1 Cascaded Framework

Detailed workflow of plant disease detection and remedy recommendation has been described in this section. First step involves Image preprocessing and leaf segmentation. These are followed by classification of diseases by multiple models. After classification comes the generation of remedies of these diseases. Everything occurs in an unified pipeline manner.

2.2 System Overview

The model of the proposed system takes a leaf image of plant as an input. Then it is processed through multiple stages of deep learning for identification of the disease. Suitable remedies are recommended thereafter. The objective lies in minimal noise in background to automate the disease detection procedure. Both convolutional and transformer based models are used for achieving maximum accuracy in classification.

2.3 Dataset Description

The experiments conducted in this study utilized the publicly available **PlantVillage dataset**, which is widely used for plant disease classification research. The dataset contains high-quality images of healthy and diseased plant leaves captured under controlled environmental conditions. It includes multiple crop species such as tomato, potato, corn, grape, apple, and others, covering a wide range of common plant diseases. The dataset consists of approximately 54,000+ labeled images belonging to 38 different classes, including both healthy and infected leaf categories. Each image is annotated with the corresponding crop type and disease label. The availability of balanced class distribution makes it suitable for supervised learning tasks. Although the dataset provides clean and well-segmented images, it is primarily collected in laboratory-controlled settings with uniform backgrounds. Therefore, additional preprocessing and augmentation techniques were applied in this study to improve the model’s robustness and simulate real-world agricultural conditions.

2.4 Image Acquisition and Preprocessing

The designed web interface collects the leaf image. These are uploaded by the user which are captured in the real world conditions including farms and fields. All images have been resized to a fixed resolution of **224 × 224 pixels**. This has been done to make sure that the trained deep learning architectures are compatible with analysing the images. Further the grayscale images are converted into RGB format. The values of the pixel are normalized using ImageNet mean and standard deviation. This has been done to maintain consistency with the transfer learning models. Data augmentation methods like random rotation, horizontal and vertical flipping, zooming, and adjusting the brightness are implemented. It improves the robustness against variations in illumination, orientation and background.

2.5 Leaf Segmentation Using U-Net

A segmentation model based on U-Net has been employed to reduce the interference in background and isolate the regions which are disease affected. Spatoal features are captured by encoder. Using the skip connections, a pixel level segmentation mask is constructed by the decoder. This architecture helpss in accurate separation of parts of leaf from the complex backgrounds. Leaf region which is segmented is extracted using the predicted mask. It is then forwarded to the classification models. The focus of models for improved performance in classification on disease patterns present on the surface of leaf has been enabled by this segmentation.

2.6 Disease Classification

A comparative multi-model approach is used to classify diseases. The deep learning architectures listed below are put into practice:

- **AlexNet:** AlexNet is an early deep convolutional neural network consisting of five convolutional layers and three fully connected layers. It uses ReLU activation and dropout for improved training performance. In this work, it served as a baseline model due to its simpler architecture and faster training time.
- **VGG16:** VGG16 is a 16-layer deep CNN that uses small 3×3 convolutional filters to learn hierarchical image features. Its uniform architecture enables stable

training and strong feature extraction. It provided consistent and high classification accuracy in disease detection.

- **VGG19:** VGG19 is a deeper extension of VGG16 with 19 weighted layers. The additional layers help capture more detailed texture and spatial features from leaf images. However, it requires higher computational resources compared to VGG16.
- **ResNet50:** ResNet50 is a 50-layer residual network that introduces skip connections to overcome the vanishing gradient problem. These residual connections allow efficient training of deeper networks. It provided stable convergence and competitive performance in classification tasks.
- **DenseNet121:** DenseNet121 connects each layer to all subsequent layers, promoting feature reuse and improved gradient flow. This dense connectivity reduces redundancy and enhances learning efficiency. It demonstrated strong generalization ability on plant disease datasets.
- **Vision Transformer (ViT):** Vision Transformer divides an image into patches and processes them using self-attention mechanisms instead of convolution. It captures global contextual relationships between image regions. ViT showed high accuracy but required greater computational resources.
- **U-Net-Assisted Classification Model:** U-Net is a segmentation model with an encoder–decoder architecture and skip connections. In this study, it was used to isolate the leaf region before classification. This preprocessing step reduced background noise and improved classification robustness.

All models are initialized with ImageNet pre-trained weights in order to apply transfer learning. The last layers of classification have been changed and fine-tuned to correspond to the categories of plant diseases. To determine attention-based global feature learning and to compare its effectiveness with convolutional neural networks, Vision Transformer is used.

2.7 Model Training and Optimization

Stratified sampling is leveraged to divide the dataset into subsets for training, validation, and testing so as to keep the class balance. Adam optimizer is applied to

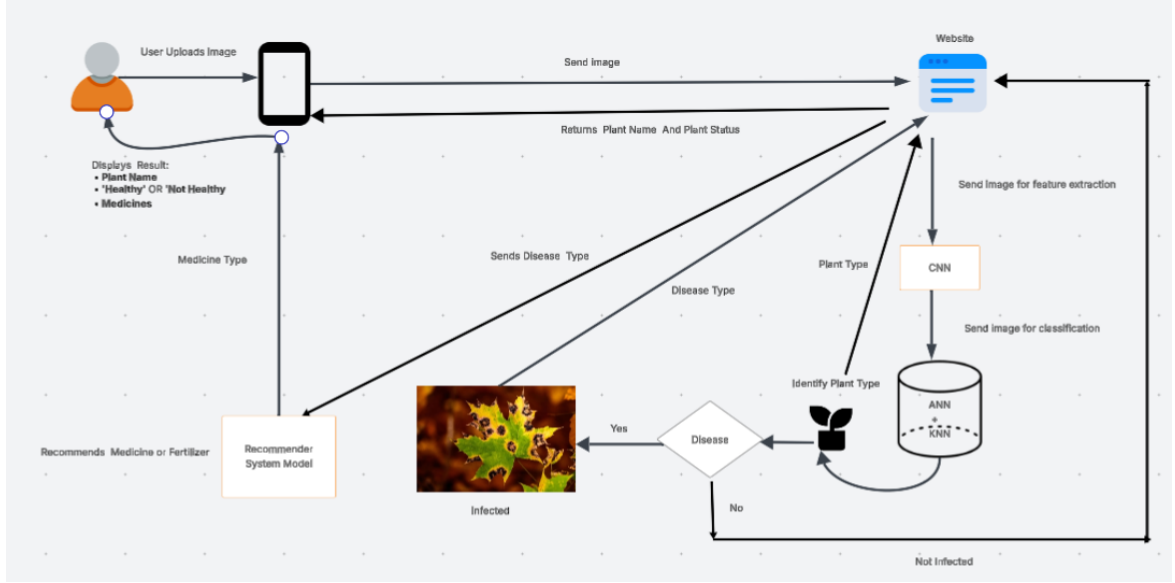


Figure 1: The Web Interface System

tune each model after the training with categorical cross-entropy loss. Learning rate scheduling and early stopping are additionally employed to prevent overfitting and ensure a stable convergence of all the models.

2.8 Remedy Recommendation Module

Next, the disease label predicted by the classifier is matched with a structured remedy database. This database holds the tailored remedy suggestions for the various diseases. The recommendations cover treatment methods, chemical or organic products, and preventive measures. The final output shown to the user is the name of the detected disease accompanied by the suggestions for the remedy.

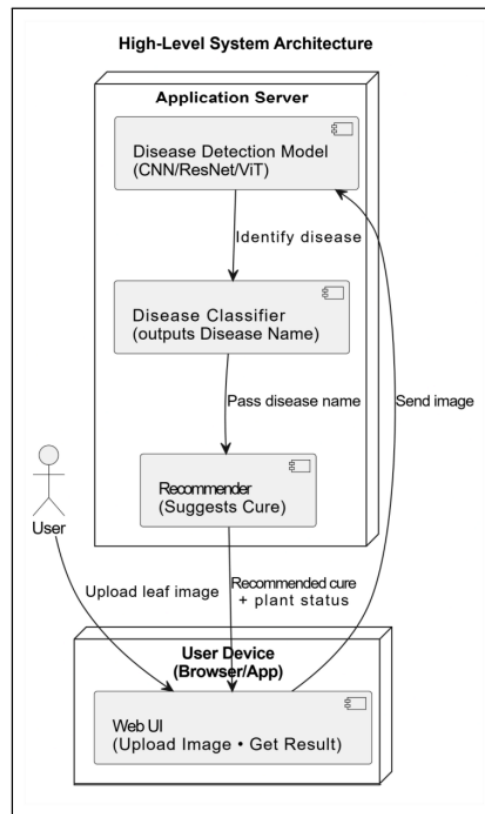


Fig. 3.2 – Client Application Diagram

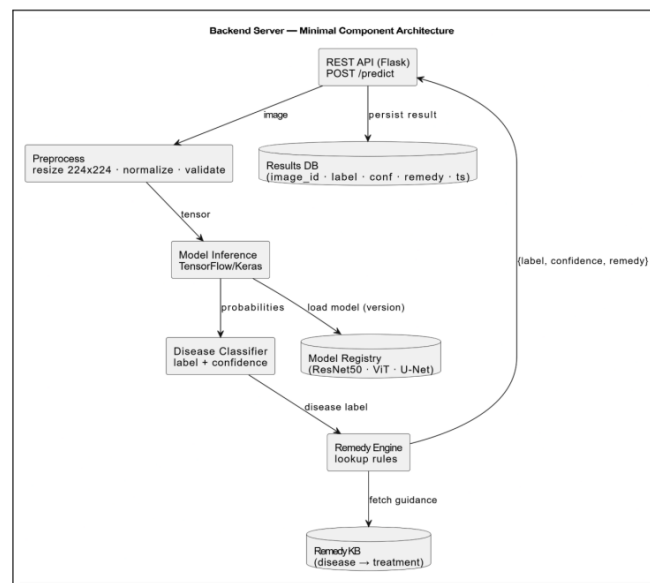


Fig. 3.3 – Backend Server Diagram

3 Experiments and Results

The experiments were carried out during the development of the project to check whether the system is able to detect plant diseases correctly. For this purpose, the PlantVillage dataset was used. This dataset contains images of healthy leaves as well as diseased leaves of different crops. The data was divided into training and testing sets so that model performance could be observed properly. Before starting the experiments, image preprocessing was done. The images were resized, normalized, and augmented. Data augmentation was necessary because the dataset otherwise led to overfitting in some models. The experiments were mainly performed to compare different deep learning models and to understand how they behave when used in a web-based application.

3.1 Experimental Setup

All the experiments were implemented using Python. TensorFlow and Keras libraries were used for training the models. Each image was resized to 224×224 pixels before being given as input to the network. During training, augmentation techniques such as rotation, flipping, zooming, and brightness changes were applied. Transfer learning was used in all models by loading ImageNet weights. Adam optimizer was used for training. Categorical cross-entropy was used as the loss function. In some cases, training was stopped early when the validation accuracy stopped improving. Learning rate was also adjusted during training when required.

3.2 Leaf Segmentation Results

Leaf segmentation was performed using the U-Net model. The main purpose of segmentation was to reduce the effect of background in the images. After segmentation, only the leaf region was considered for classification. During testing, it was seen that U-Net was able to extract the leaf region in most images. However, in a few cases with complex background, the segmentation was not perfect. Even then, the segmented images helped the classification models to focus more on disease patterns.

3.3 Classification Performance

Several deep learning models were trained during the project. These included AlexNet, VGG16, VGG19, ResNet50, DenseNet121, and Vision Transformer. All models were trained on the same dataset so that comparison could be done fairly. The performance was evaluated mainly using training and testing accuracy. VGG16 gave the most consistent results among all models. Its training was stable and the accuracy obtained was high. VGG19 and DenseNet121 gave good results, but both of them took more time to train compared to other models. ResNet50 was easier to train and gave stable results during experimentation. AlexNet trained quickly, but its accuracy was lower than the other models. The Vision Transformer (ViT) model also showed good performance in testing based on reaching one of the highest classification accuracy among all the models that were tested. However, it has relatively lower training accuracy and greater computational demand, which suggests that transformer-based architectures can be trained only with great attention to hyperparameters and longer training times in order to reach the full representational potential.

3.4 Web-Based System Evaluation

After the models were trained, they were connected to a web-based system. Flask was used to handle the backend part of the application. A frontend was created where users can upload images of leaf. While testing the system, image uploads and predictions worked correctly in most cases. The response time was reasonable, and the segmentation and classification steps worked together without major problems. Some practical issues were faced during integration, such as handling image size and response time, but these were resolved during testing.

Table 2: Model Results

Dataset	VGG-16	VGG-19	ResNet	DenseNet	ViT	MobileNetV	AlexNet	U-Net
Training Accuracy	0.9820	0.9800	0.680		0.9995	0.9230	0.94	0.7820
Testing Accuracy	0.9840	0.9800	0.634	0.1053	0.9978	0.8994	0.98	0.7250
External Validation	0.1282	0.1650	–	–	–	–	–	–

3.5 Result Analysis

The comparative evaluation of different deep learning models highlights several important observations. Among the convolutional neural network architectures, VGG16

achieved the most consistent and stable performance across training and testing datasets. Its balanced depth and uniform architecture enabled effective extraction of disease-related texture features from leaf images. VGG19 and DenseNet121 also demonstrated competitive performance; however, their higher computational complexity resulted in longer training times. ResNet50 showed stable convergence due to residual connections, but its testing accuracy was comparatively lower, which may indicate sensitivity to dataset characteristics. The Vision Transformer (ViT) achieved one of the highest testing accuracies, indicating its capability to capture global contextual relationships using self-attention mechanisms. However, its relatively lower training accuracy and higher computational demand suggest that transformer-based models require careful hyperparameter tuning and larger datasets to fully utilize their potential. AlexNet trained faster due to its simpler architecture but showed comparatively lower accuracy, suggesting that deeper architectures are more effective in learning complex disease patterns. The integration of U-Net-based segmentation improved classification robustness by reducing background interference and allowing models to focus primarily on infected leaf regions. The external validation results indicate a significant drop in performance for some models. This highlights a common limitation of using controlled datasets such as PlantVillage, as models may struggle when exposed to real-world field conditions. Therefore, improving dataset diversity and incorporating real-field images remain important directions for future work.

4 Conclusion

This paper demonstrates a detailed plant disease recognition and recovery prescription system on the basis of a comparative analysis of seven deep learning systems, including U-Net, ResNet50, Vision Transformer (ViT), VGG16, VGG19, AlexNet, and DenseNet121. Through the combination of the convolution-based and transformer-based model, the proposed framework proves to be robust to various types of plant diseases and image variations. One of the contributions is the use of U-Net-based segmentation to isolate the leaf areas against the complicated backgrounds so that the models can concentrate on disease-specific visual information and boost prediction quality, especially where the disease signs can be too young or too faint. There are experimental findings that transformer-based models (ViT) are able to learn complex texture patterns, and ResNet and DenseNet models offer stable and competitive training, and validation performance. Besides disease classification, the system also has the remedy recommendation module where the predicted diseases are mapped to the relevant organic control, chemical control, preventive and good agricultural practices, thus the framework is converted into an effective decision-support tool to the farmers. Although these yield promising results, these have constraints because of the dependence on the PlantVillage dataset that was gathered in controlled settings, there is a problem of class imbalance and the large models that consume high computational power which may not be achievable in low-resource devices. These difficulties indicate the necessity of the future validation in terms of real farm pictures and streamlined lightweight models.

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